

Time Series — Solutions

Practice Exam — Week 11: Diagnostics, Model Selection & Long Memory

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Problem 1 (5 points)

Statement.

(Residual diagnostics and the danger of R^2 .) A correctly specified, stationary AR(1) model $X_t = \phi X_{t-1} + \varepsilon_t$ ($|\phi| < 1$) is fitted with the true parameters known. Recall that the one-step fitted value is $\hat{X}_t = \phi X_{t-1}$, the residual is $e_t = X_t - \hat{X}_t$, and that for white-noise residuals each sample residual autocorrelation $\hat{r}_\tau$ is approximately $N(0, 1/n)$.

(a) Show that the residual equals the innovation, $e_t = \varepsilon_t$, and use this to state which of the four standard residual checks (constant zero mean, constant variance, no autocorrelation, Gaussianity) the residuals must pass and why. (1.5 pts)

(b) For $n = 400$ observations, give the numerical pointwise 95% correlogram band $(-1.96/\sqrt{n}, 1.96/\sqrt{n})$ for the residual ACF, and explain in one sentence why eyeballing this band lag-by-lag is not a substitute for a portmanteau test. (1.5 pts)

(c) Show that for an AR(1) the population $R^2 = 1 - s_e^2/s_y^2$ equals ϕ^2 , where s_e^2 is the residual variance and $s_y^2 = \text{Var}(X_t)$. Evaluate it at $\phi = 0.6$, and explain why a “low” value here is no indictment of the model and why a higher R^2 obtained by fitting an AR(4) to the same data is no endorsement. (2 pts)

Solution.

(a) Substitute the model equation into the residual: with $\hat{X}_t = \phi X_{t-1}$,

$$e_t = X_t - \hat{X}_t = (\phi X_{t-1} + \varepsilon_t) - \phi X_{t-1} = \varepsilon_t.$$

The AR structure cancels exactly, leaving the innovation. Consequently the residuals *are* the white noise, so all four checks must pass: **(i) constant zero mean** because $\mathbb{E}[\varepsilon_t] = 0$ for all t ; **(ii) constant variance** (homoscedasticity) because $\text{Var}(\varepsilon_t) = \sigma^2$ for all t ; **(iii) no autocorrelation** because $\text{Cov}(\varepsilon_t, \varepsilon_s) = 0$ for $t \neq s$ (white noise); **(iv) Gaussianity** whenever the innovations ε_t are Gaussian (this is the only check that is *conditional* on a distributional assumption on ε_t rather than automatic). This is the template for all residual diagnostics: any departure from these four properties is evidence *against* the fitted model.

(b) With $n = 400$, $\sqrt{n} = 20$, so the band is

$$\left(-\frac{1.96}{\sqrt{400}}, \frac{1.96}{\sqrt{400}}\right) = \left(-\frac{1.96}{20}, \frac{1.96}{20}\right) = (-0.098, 0.098).$$

Eyeballing this ± 0.098 band lag-by-lag is a *pointwise* 95% test for each single $\hat{r}_\tau$; checking many lags simultaneously creates a multiple-testing problem (under the null, about 5% of the lags will breach the band by chance), so a single *portmanteau* statistic that aggregates many lags at once (Box–Pierce / Ljung–Box) is needed to control the overall error rate.

(c) For an AR(1) the marginal (process) variance is the geometric sum of the squared MA(∞) weights ϕ^{2j} ,

$$s_y^2 = \text{Var}(X_t) = \sigma^2 \sum_{j=0}^{\infty} \phi^{2j} = \frac{\sigma^2}{1 - \phi^2},$$

while the one-step residual variance is $s_e^2 = \text{Var}(e_t) = \text{Var}(\varepsilon_t) = \sigma^2$ by part (a). Hence

$$R^2 = 1 - \frac{s_e^2}{s_y^2} = 1 - \frac{\sigma^2}{\sigma^2/(1 - \phi^2)} = 1 - (1 - \phi^2) = \phi^2.$$

At $\phi = 0.6$: $R^2 = 0.6^2 = 0.36$. A value of 0.36 “looks low” but it is in fact the *best attainable*: a stochastic process is not perfectly predictable, so even the true, optimal model leaves the fraction $1 - \phi^2$ of the variance unexplained — a low R^2 is therefore no indictment. Conversely, fitting a richer AR(4) to the same data can only *increase* the in-sample R^2 (adding parameters never reduces explained variance), so a higher R^2 is no endorsement of the larger model. The correct tool for choosing between AR(1) and AR(4) is an information criterion (AICc/BIC) or a Ljung–Box test on the AR(1) residuals, *not* R^2 .

Problem 2 (5 points)

Statement.

(Information criteria: AIC, AICc, BIC.) For a model with k estimated parameters fitted to n observations with maximised likelihood L ,

$$\text{AIC} = -2 \ln L + 2k, \quad \text{AICc} = -2 \ln L + 2k \frac{n}{n - k - 1}, \quad \text{BIC} = -2 \ln L + k \ln n,$$

and for an ARMA(p, q) the parameter count is $k = p + q + 1$. Smaller is better for all three.

(a) Prove the identity

$$\text{AICc} - \text{AIC} = \frac{2k(k+1)}{n - k - 1},$$

and state precisely the conditions on n and k under which AIC and AICc become approximately equivalent. (1.5 pts)

(b) Consider a mean-zero ARMA(p, q) model in which exactly m of the $p + q$ AR and MA coefficients are known in advance and fixed to zero (only the remaining ones, and the innovation variance, are estimated). Give the estimated-parameter count k and write down the resulting AICc in closed form. (1.5 pts)

(c) A full AR(4) model ($k = p + q + 1 = 5$) is compared with a constrained AR(4) in which $m = 2$ coefficients are fixed to zero, on $n = 30$ observations. Compute the AICc complexity penalty $2k n / (n - k - 1)$ for each model, and state by how much the constrained model’s $-2 \ln L$ is allowed to exceed the full model’s before AICc would prefer the full model. (2 pts)

Solution.

(a) Subtract the two criteria and put the bracket over a common denominator:

$$\begin{aligned} \text{AICc} - \text{AIC} &= 2k \left(\frac{n}{n - k - 1} - 1 \right) = 2k \cdot \frac{n - (n - k - 1)}{n - k - 1} \\ &= 2k \cdot \frac{k + 1}{n - k - 1} = \frac{2k(k + 1)}{n - k - 1} (> 0). \end{aligned}$$

The correction is always positive (AICc penalises complexity more heavily). The two criteria are approximately equivalent exactly when this term is small:

- if k is **fixed** and $n \rightarrow \infty$, then $\frac{2k(k+1)}{n - k - 1} \rightarrow 0$, so $\text{AICc} \rightarrow \text{AIC}$;
- if $k = k_n$ grows with n , the gap is still negligible provided n is large relative to k_n^2 , e.g. $k_n^2/n \rightarrow 0$. If k grows too fast ($k_n^2/n \not\rightarrow 0$) the gap need not vanish and AICc stays meaningfully larger.

In words: *large samples relative to the number of parameters* make the two agree; small samples or richly parameterised models keep AICc strictly more conservative.

(b) If m of the $p + q$ AR/MA coefficients are fixed (not estimated), only $p + q - m$ of them are estimated. Adding the innovation variance σ^2 (and, for a mean-zero model, no separate mean parameter), the estimated-parameter count is

$$k = p + q + 1 - m.$$

Substituting into the AICc formula,

$$\text{AICc}(\theta) = -2 \ln L(\theta | y) + 2(p + q + 1 - m) \frac{n}{n - (p + q + 1 - m) - 1}.$$

Fixing coefficients to zero lowers k , which both shrinks the penalty and (in a subsequent Ljung–Box test) raises the residual degrees of freedom.

(c) With $n = 30$:

- Full AR(4): $k = p + q + 1 = 4 + 0 + 1 = 5$, so

$$2k \frac{n}{n - k - 1} = 2 \cdot 5 \cdot \frac{30}{30 - 5 - 1} = 10 \cdot \frac{30}{24} = 12.5.$$

- Constrained AR(4), $m = 2$ fixed: $k = 4 + 0 + 1 - 2 = 3$, so

$$2k \frac{n}{n - k - 1} = 2 \cdot 3 \cdot \frac{30}{30 - 3 - 1} = 6 \cdot \frac{30}{26} = \frac{180}{26} \approx 6.923.$$

The constrained model carries the smaller penalty by $12.5 - 6.923 = 5.577$. Writing $\text{AICc} = -2 \ln L + \text{penalty}$, AICc prefers the full model only when

$$[-2 \ln L_{\text{full}} + 12.5] < [-2 \ln L_{\text{constr}} + 6.923],$$

i.e. when $-2 \ln L_{\text{constr}} - (-2 \ln L_{\text{full}}) > 12.5 - 6.923 = 5.577$. So the constrained model's $-2 \ln L$ may exceed the full model's by up to $\boxed{\approx 5.58}$ before AICc switches its preference to the full AR(4); equivalently, the two extra estimated coefficients must improve the log-likelihood by more than ≈ 2.79 to be worth keeping under AICc.

Problem 3 (5 points)

Statement.

(Portmanteau tests and the degrees of freedom.) For $n = 200$ observations a mean-zero ARMA(2, 1) model is fitted, and the residual sample autocorrelations at lags 1 to 5 are

$$\hat{r}_1 = 0.05, \quad \hat{r}_2 = -0.08, \quad \hat{r}_3 = 0.12, \quad \hat{r}_4 = 0.04, \quad \hat{r}_5 = -0.10.$$

Recall the Box–Pierce statistic $Q_m^{\text{BP}} = n \sum_{\tau=1}^m \hat{r}_\tau^2$ and the Ljung–Box statistic $Q_m = n(n+2) \sum_{\tau=1}^m \hat{r}_\tau^2 / (n - \tau)$, both approximately χ_{m-p-q}^2 under the null that the residuals are white up to lag m . Useful upper-5% quantiles: $\chi_{2,0.95}^2 = 5.991$, $\chi_{3,0.95}^2 = 7.815$, $\chi_{4,0.95}^2 = 9.488$.

(a) Compute the Ljung–Box statistic Q_5 (show the term-by-term sum). (2 pts)

(b) Test $H_0 : \rho_1 = \dots = \rho_5 = 0$ at the 5% level for the ARMA(2, 1) fit: state the degrees of freedom, the critical value, and the conclusion. (1.5 pts)

(c) Suppose the same residual autocorrelations had instead come from an AR(1) fit. Redo the test (state the new df , critical value, conclusion) and explain, in one sentence, why getting the degrees of freedom right ($df = m - p - q$, not m) is essential. (1.5 pts)

Solution.

(a) With $n = 200$ we have $n(n+2) = 200 \cdot 202 = 40400$, and $m = 5$. The Ljung–Box statistic is

$$Q_5 = 40400 \left(\frac{0.05^2}{199} + \frac{(-0.08)^2}{198} + \frac{0.12^2}{197} + \frac{0.04^2}{196} + \frac{(-0.10)^2}{195} \right).$$

Term by term,

$$\begin{aligned} \frac{0.0025}{199} &= 1.256 \times 10^{-5}, & \frac{0.0064}{198} &= 3.232 \times 10^{-5}, & \frac{0.0144}{197} &= 7.310 \times 10^{-5}, \\ \frac{0.0016}{196} &= 8.163 \times 10^{-6}, & \frac{0.0100}{195} &= 5.128 \times 10^{-5}. \end{aligned}$$

Summing the five terms gives

$$1.256 \times 10^{-5} + 3.232 \times 10^{-5} + 7.310 \times 10^{-5} + 8.163 \times 10^{-6} + 5.128 \times 10^{-5} = 1.7743 \times 10^{-4}.$$

Hence

$$Q_5 = 40400 \times 1.7743 \times 10^{-4} \approx 7.17.$$

(b) For the ARMA(2, 1) fit, $p = 2$ and $q = 1$, so the degrees of freedom are

$$\text{df} = m - p - q = 5 - 2 - 1 = 2.$$

The 5% critical value is $\chi_{2,0.95}^2 = 5.991$. Since $Q_5 \approx 7.17 > 5.991$, we **reject** H_0 at the 5% level: there is significant residual autocorrelation up to lag 5, so the ARMA(2, 1) model is *inadequate* and should be revised.

(c) For an AR(1) fit, $p = 1$, $q = 0$, so

$$\text{df} = m - p - q = 5 - 1 - 0 = 4,$$

and the critical value rises to $\chi_{4,0.95}^2 = 9.488$. Now $Q_5 \approx 7.17 < 9.488$, so we would **fail to reject** H_0 (the test deems the residuals adequately white). The *same* numbers thus give opposite conclusions purely through the degrees of freedom: subtracting the correct number of estimated ARMA parameters ($p + q$) is essential, because using $\text{df} = m$ would inflate the critical value (here $\chi_{5,0.95}^2 = 11.07$) and make the test reject far too rarely, masking genuine misspecification.

Problem 4 (5 points)

Statement.

(Revision of Week 10: forecasting a Gaussian AR(1).) Let $X_t = \phi X_{t-1} + \varepsilon_t$ be a causal Gaussian AR(1) with $\varepsilon_t \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2)$, and recall that for a causal ARMA process the h -step forecast error variance is $\text{Var}(e_n(h)) = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2$, where $\{\psi_j\}$ are the MA(∞) weights.

(a) Give the MA(∞) weights ψ_j of the AR(1) and, using that the best linear predictor zeroes future innovations, show that $X_{n+h}^n = \phi^h X_n$. (1.5 pts)

(b) Show that $\text{Var}(e_n(h)) = \sigma^2 \frac{1 - \phi^{2h}}{1 - \phi^2}$ and verify that, as $h \rightarrow \infty$, both the point forecast and the error variance converge to their stationary limits (the mean 0 and the marginal variance γ_0 respectively); identify γ_0 . (1.5 pts)

(c) Take $\phi = 0.6$, $\sigma^2 = 4$ and the last observation $X_n = 5$. Compute the two-step forecast X_{n+2}^n , its forecast-error variance, and a 95% Gaussian prediction interval for X_{n+2} (use $z_{0.975} = 1.96$, $\sqrt{5.44} \approx 2.332$). (2 pts)

Solution.

(a) Iterating the causal AR(1) recursion gives the MA(∞) representation $X_t = \sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j}$, so the weights are

$$\psi_j = \phi^j, \quad j = 0, 1, 2, \dots$$

For the best linear (here, since Gaussian, also overall best) predictor, write $X_{n+h} = \phi^h X_n + \sum_{j=1}^h \phi^{h-j} \varepsilon_{n+j}$ and take the conditional expectation given $X_1, \dots, X_n$. Every future innovation $\varepsilon_{n+1}, \dots, \varepsilon_{n+h}$ has conditional expectation 0 (it is uncorrelated with the data, and zero-mean), leaving

$$X_{n+h}^n = \mathbb{E}[X_{n+h} \mid X_1, \dots, X_n] = \phi^h X_n.$$

(b) The h -step forecast error is $e_n(h) = X_{n+h} - X_{n+h}^n = \sum_{j=0}^{h-1} \psi_j \varepsilon_{n+h-j} = \sum_{j=0}^{h-1} \phi^j \varepsilon_{n+h-j}$, a sum of uncorrelated terms, so

$$\text{Var}(e_n(h)) = \sigma^2 \sum_{j=0}^{h-1} \psi_j^2 = \sigma^2 \sum_{j=0}^{h-1} \phi^{2j} = \sigma^2 \frac{1 - \phi^{2h}}{1 - \phi^2}.$$

As $h \rightarrow \infty$, since $|\phi| < 1$ we have $\phi^h \rightarrow 0$, so the point forecast $X_{n+h}^n = \phi^h X_n \rightarrow 0$, the process mean; and $\phi^{2h} \rightarrow 0$, so

$$\text{Var}(e_n(h)) \rightarrow \frac{\sigma^2}{1 - \phi^2} = \gamma_0,$$

the marginal variance of the AR(1). Far-ahead forecasting thus reverts to “the unconditional distribution”: point forecast $\rightarrow$ mean, error variance $\rightarrow \gamma_0$.

(c) With $\phi = 0.6$, $\sigma^2 = 4$, $X_n = 5$:

$$X_{n+2}^n = \phi^2 X_n = 0.36 \times 5 = 1.8.$$

The two-step forecast-error variance is

$$\text{Var}(e_n(2)) = \sigma^2(1 + \phi^2) = 4(1 + 0.36) = 4 \times 1.36 = 5.44,$$

so the standard error is $\sqrt{5.44} \approx 2.332$. A 95% Gaussian prediction interval for X_{n+2} is

$$X_{n+2}^n \pm z_{0.975} \sqrt{\text{Var}(e_n(2))} = 1.8 \pm 1.96 \times 2.332 = 1.8 \pm 4.571,$$

i.e.

$$\boxed{X_{n+2} \in (-2.77, 6.37) \text{ with 95\% confidence.}}$$

(As a check, $\gamma_0 = \sigma^2/(1 - \phi^2) = 4/0.64 = 6.25$, and indeed $\text{Var}(e_n(2)) = 5.44 < 6.25$, with the gap closing as h grows.)